

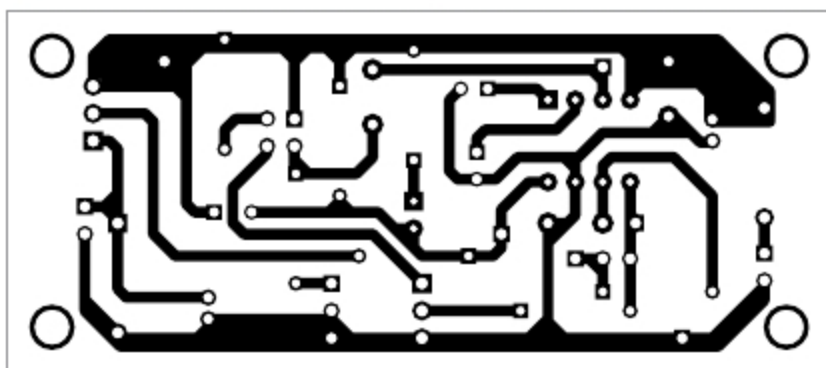


Fig. 3: Actual-size PCB layout for the automatic water dispenser

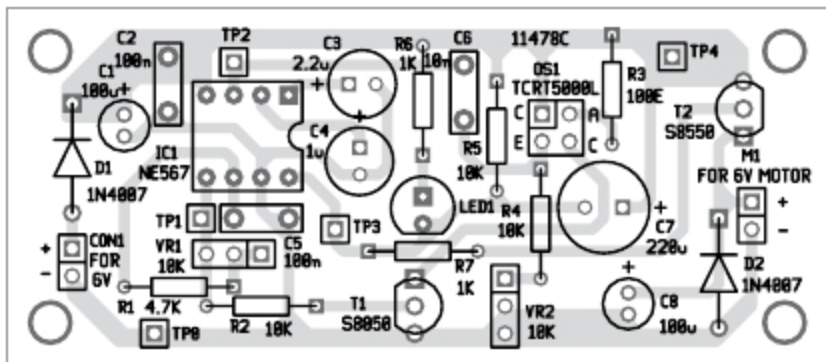


Fig. 4: Components layout of the PCB

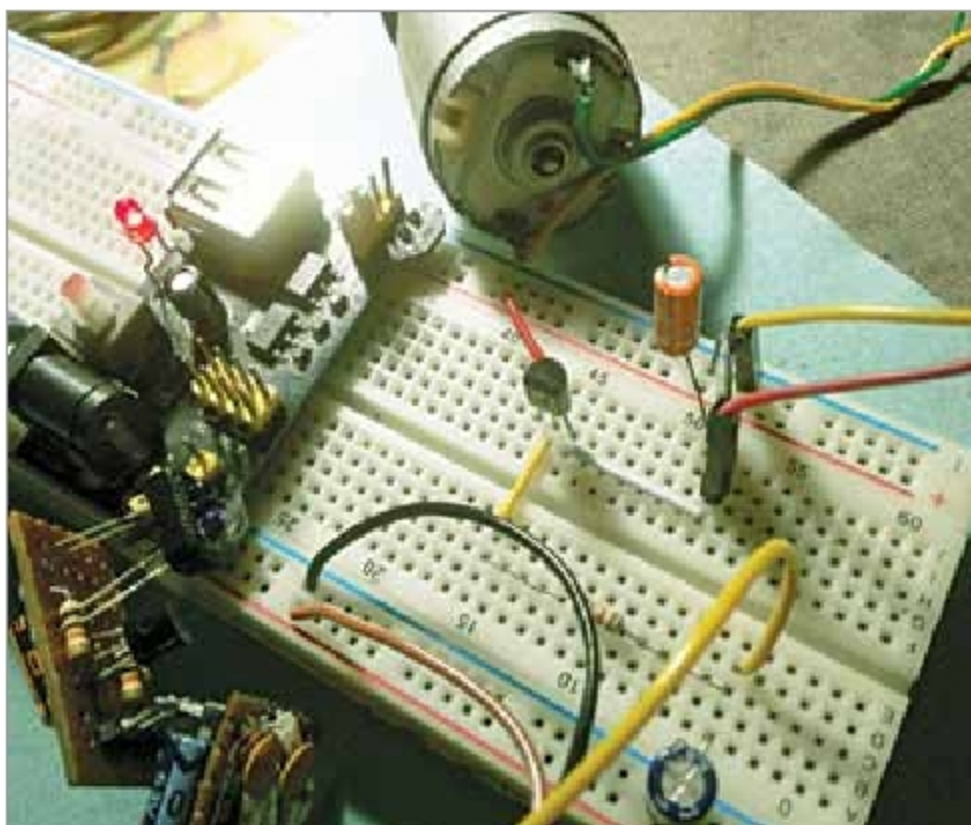


Fig. 6: Author's prototype of pump driver

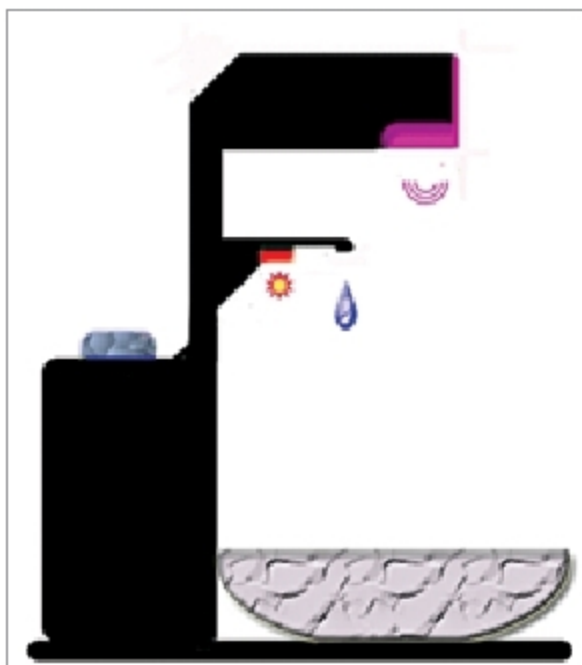


Fig. 5: Proposed arrangement of the automatic water dispenser

silicone hose to the motor outlet, and submerge it in water to run it. Make sure that the water level is always higher than the motor as dry run will damage the water pump motor in a flash. Fortunately, we can prevent such a mishap by adding a mini float switch to the wiring of the water pump.

Here, S8550 pnp transistor (T2) is used as a high-side switch to drive the water pump motor (M1). Since the transistor switches the high voltage on and off (instead of ground), the setup provides a good return path when the water pump motor is turned off, and hence there's not much undesired electromagnetic interference (EMI).

Further, when the output of IC1 goes high (idle state), the base resistor

(R7) of T2 is floating. So, the V_{BE} of T2 becomes 0V ($5.3 - 5.3 = 0$) and keeps T2 and M1 dead off.

Nevertheless, the driver circuitry can be replaced with a low-current 5V relay because IC1 can merrily source current up to 100mA.

Circuit and working

The circuit diagram of the automatic water dispenser is shown in Fig. 2. It is built around tone decoder NE567 (IC1), optical reflective sensor TCRT5000L (OS1), two 1N4007 rectifier diodes (D1 and D2), two transistors S8050 and S8550 (T1 and T2), and a few other components.

The working of the circuit is simple. Initially, when there is no object such as a hand or pet in front of TCRT5000, LED1 remains off and no water flows out as the water pump also remains off. But when there is an object/obstacle in front of TCRT5000, LED1 glows and, at the same time, the water pump is switched on and water flows out from the tap.

Construction and testing

Wiring up the circuit is simple. You can either use a veroboard, or the designed PCB layout given here. An actual-size PCB layout for the automatic water dispenser is shown in Fig. 3 and its components layout in Fig. 4.

After assembling the circuit on the PCB, enclose it in a water-resistant (transparent/translucent) prototype

enclosure. A status indicator (LED1) has been added in the circuit, which glows when the proximity sensor detects an object/obstacle. Fix the TCRT5000L sensor and LED1 on the front panel of the cabinet.

A trimpot (VR1) is used to set the operating frequency of the oscillator. The power supply used here is 6V (four AA/AAA cells) DC supply. It also powers the mini submersible water pump (M1). The final proposed arrangement is shown in Fig. 5. For troubleshooting, check voltages at various test points as per the table.

Note. The system hardly needs any crucial electrical adjustments. The operating frequency of NE567 (IC1) can be tuned in 1kHz to 2kHz range using a 10-kilo-ohm potmeter (VR1). You can check frequency at test point TP1.

In case of valid proximity detection, test point TP2 gives a value greater than 150mV. The operating distance lies in 7cm to 15cm range depending upon the setting of the second 10-kilo-ohm trimpot (VR2).

The primary output pin 8 of NE567 is actually a free-floating output transistor collector. It gives a logic-low level output (TP3) with 100mA current-sinking capability on proximity detection. **EFY**



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